Pu Xia

PERSONAL INFORMATION
Address: Edgbaston, Birmingham, B15 2TT, UK
Tel: +44 7770748713
Email: p.xia@bham.ac.uk
Google Scholar: https://scholar.google.com/citations?user=D41a8SkAAAAJ&hl=zh-CN
ResearchGate: https://www.researchgate.net/profile/Pu-Xia-4
University:
https://www.birmingham.ac.uk/schools/biosciences/staff/profile.aspx?ReferenceId=204848&Name=dr-pu-xia

EDUCATION

2012-2018	Ph.D. in Environmental Science, Nanjing University, China
2008-2012	B.Sc. in Environmental Science, Nanjing University, China

POSITIONS

Since 2023	Assistant Professor, School of Bioscience, University of Birmingham, UK
2022-2023	Postdoctoral Fellow, Toxicology Centre, University of Saskatchewan, CA (Supervisor: John Giesy)
2019-2022	Postdoctoral Fellow, Environmental and Climate Change Canada, CA (Supervisors: Doug Crump, Jason M O'Brien)
2019-2022	Postdoctoral Fellow, Department of Biology, University of Ottawa, CA (Supervisor: Laurie Chan)

TEACHING EXPERIENCE

Since 2023	Module Lead for Master Project: New approach methodologies for regulatory toxicology, University of Birmingham, UK
2023-2025	EMBL-EBI: Gene-environment interactions in human health and disease
2016-2017	Teaching Assistant in Ecotoxicology Advance, Nanjing University, China
2015-2016	Teaching Assistant in Ecotoxicology Advance, Nanjing University, China

Academic Service Roles

- Editorial Board Members for npj Emerging Contaminants (https://www.nature.com/npjemergcontam/editors)
- Guest Associate Editor for Toxics (Section: Ecotoxicology of Pollutants of High Concern)

- Guest Associate Editor for Frontiers in Environmental Science (Special Issues: Modeling for environmental pollution and change)
- Guest Associate Editor for Frontiers in Endocrinology (Section: Translational and Clinical Endocrinology)
- Guest Associate Editor for Frontier in Public Health (Section: Environmental Health and Exposome)
- ASPIS quantitative AOP Working Group Co-chair (https://aspis-cluster.eu/working-groups/)

Served as a peer reviewer for international academic journals in the field of environmental science and toxicology, including ES&T, JHM, Aquatic Toxicology, Ecotoxicology and Environmental Safety, ET&C, ACS Omega, Nucleic Acids Research, Marine Pollution Bulletin, Cell Biology and Toxicology, Toxics, Computational and Structural Biotechnology Journal, International Journal of Environmental Research and Public Health, Environmental Science and Ecotechnology, and Environment & Health.

GRANT SUPPORT

PI or Co-I

2026-2028	**PI**, CERJ-PARC University-Industry Collaboration with ExxonMobil Biomedical Sciences Inc (EMBSI). **£90k**
2025-2029	**PI**, BBSRC collaborative training partnerships (CTP) scheme PhD studentship with Unilever. **£155k**
2025-2029	**PI**, EPSRC Industrial Doctoral Landscape Award (IDLA) PhD studentship with Unilever. **£168k**
2024-2026	**Co-I,** BBSRC IAA - Longview: Developing, validating, and enhancing, high-throughput monitoring systems for Daphnia Magna. **£17.5k**
2024-2024	**PI**, University of Birmingham Internal funding - Interdisciplinary Research (IDR) Accelerator Award (2025). **£10k**
2024-2024	**PI**, University of Birmingham Pumping Prime Award. **£5k**
2023-2025	**PI**, University of Birmingham Start-up grant. **£30k**
2017-2018	**PI**, Program B for Outstanding Ph.D. Candidates, Nanjing University, Screening and Prediction of Chemical Toxicity Using Functional Genomics Approaches. **¥40k**

Participant

2022-2029	Horizon Europe, European Partnership for the Assessment of Risks from Chemicals (PARC). **£7M**

2022-2029	Horizon Europe, European Partnership for the Assessment of Risks from Chemicals (PARC). **£7M**
2021-2026	H2020, PrecisionTox: New Approach Methodologies for Chemical Safety. **£19M**
2019-2023	Natural Science and Engineering Research Council of Canada: Monitoring of SARS-CoV-2 Variants by RNA in Municipal Wastewater (#326415-07) **CA$1M**
2016-2021	EcoToxChip: Toxicogenomics assessment of wildlife birds exposed to environmental pollutants. **CA$9.6M**
2013-2018	European FP7 SOLUTIONS: Solutions for present and future emerging pollutants in land and water resources management **€120k**

AWARDS

2022 USask's 2022 Publicly Engaged Scholarship Team Award, Canada
2020 The 100th Outstanding Doctoral Dissertation Award in Jiangsu Province
2020 Outstanding Doctoral Dissertation Award of Nanjing University
2018 ORGANO Doctoral Research Awards (First Prize), Organo Corporation ($2000)
2017 Pillar Scholarship from Nanjing University ($1000)
2017 Gao Tingyao Talent Scholarship for Outstanding Young Doctor ($6000)
2017 Program B for Outstanding Ph.D. Candidates from Nanjing University ($8000)
2017 Asia Pacific SETAC Best Student Oral Presentation Award ($100)
2017 Environmental Science & Technology Best Papers of 2016 (Second Runner-Up)
2017 Quan Xin Environmental Innovation Scholarship, Nanjing University
2016 ACS Editor's Choice from American Chemical Society
2016 Excellent Graduate Honor from Nanjing University
2016 National Scholarship for Doctoral Students from Nanjing University ($5000)
2014 Student travel award from Jyvaskyla University, Finland, 2014 ($2000)

PEER REVIEWED PUBLICATIONS

(*corresponding author；#co-first author)

1. **Xia P**, Wang P, Liu S, Peng Y, Gou X, Tan H, Guo J, Shi W, Colbourne JK, Giesy JP, Zhang X. Network-Based Assessment of Environmental Neurotoxicants Using High-Throughput Transcriptomics in Zebrafish Embryos. ***Environ. Sci. Technol.*** 2026, 60, 11, 8326–8338 (Supplementary Cover)
2. Wang P, **Xia P**, Gao S, et al. Critical Structures of Bisphenol Analogues on Embryonic Toxicity Identified by a Computational Approach. ***Environ. Sci. Technol.*** 2025 Jan 28;59(3):1553-1564.
3. Tian M, Ding X, Pang Y, Xu D, Sun Y, **Xia P**. Transcriptomic dose-response by UVC and heavy ion radiation reveal pathways to immune impairment. *Toxicol In Vitro.* 2025 May 24;108:106086.

4. Ankley PJ, Challis J, **Xia P**, Gong Y, Zhou Y, Hecker M, Giesy JP, Brinkmann M. Interactions of erythromycin and an antibiotic mixture with the gut microbiome of juvenile rainbow trout. *Chemosphere*. 2025 May;377:144263.
5. Wang P, **Xia P**, Gao S, Shi W, Zhang X. Critical Structures of Bisphenol Analogues on Embryonic Toxicity Identified by a Computational Approach. *Environ Sci Technol.* 2025 Jan 28;59(3):1553-1564. 2025 Jan 10.
6. Wu Q, Wang X, **Xia P**, Zhang X, Zhang R. Editorial: Modeling for Environmental Pollution and Change. *Frontiers in Environmental Science.* 26 Sep 2024.
7. Wu Q, Liu H, Zhang R, Zhang X, **Xia P***. Environmental endocrine disruptors: mechanisms, implications, and advances in detection and mitigation in endocrinology. *Frontiers in Endocrinology*. *15*, 1510917. 06 Nov 2024.
8. Guo J, Luo Y, Fang C, Jin J, **Xia P**, Wu B, Zhang X, Yu H, Ren H, Shi W. Advancing the Effect-Directed Identification in Combined Pollution: Using Pathways to Link Effects and Toxicants. *Environmental Science & Technology*. 2024 Oct 11.
9. Gou X, Tian M, Yan L, **Xia P**, Ji H, Tan H, Shi W, Yu H, Zhang X. A novel molecular pathway of lipid accumulation in human hepatocytes caused by PFOA and PFOS. *Environment international*. 2024 Sep 1;191:108962.
10. Asadi M, Hamilton D, Shomachuk C, Oloye FF, De Lange C, Liang J, **Xia P**, Osunla CA, Cantin J, Mejia EM, Gregorchuk BS. Assessment of Rapid and Conventional RT-qPCR-Based Systems for Wastewater Surveillance. *ACS ES&T Water*. 2024 Sep 5;4(10):4333-42.
11. Oloye FF, Xie Y, Asadi M, Challis JK, Osunla CA, **Xia P**, Cantin J, Femi-Oloye OP, Brinkmann M, McPhedran KN, Sadowski M, Pandey S, Jones PD, Mangat C, Servos MR, Giesy JP. Solid-liquid distribution of SARS-CoV-2 in primary effluent of a wastewater treatment plant. *MethodsX*. 2024 Mar 11;12:102645.
12. Oloye FF, Asadi M, Yusuf W, Champredon D, **Xia P**, Femi-Oloye OP, De Lange C, El-Baroudy S, Osunla CA, Xie Y, Cantin J. Rapid Spread of Omicron Sub-Lineage as Evidence by Wastewater Surveillance. *Water.* 2024 Jan;16(2):318.
13. Champredon D, Becker D, Peterson SW, Mejia E, Hizon N, Schertzer A, Djebli M, Oloye FF, Xie Y, Asadi M, Cantin J, **Xia P**, Osunla CA, Brinkmann M, McPhedran KN, Servos MR, Giesy JP, Mangat C. Emergence and spread of SARS-CoV-2 variants of concern in Canada: a retrospective analysis from clinical and wastewater data. BMC Infect Dis. 2024 Jan 29;24(1):139.
14. Tan H, Gao P, Luo Y, Gou X, **Xia P**, Wang P, Yan L, Zhang S, Guo J, Zhang X, Yu H. Are New Phthalate Ester Substitutes Safer than Traditional DBP and DiBP? Comparative Endocrine-Disrupting Analyses on Zebrafish Using In Vivo, Transcriptome, and In Silico Approaches. *Environ. Sci. Technol.* 2023, 57, 37, 13744–13756.
15. Asadi M, Hamilton D, Shomachuk C, Oloye FF, De Lange C, **Xia P**, Osunla CA, Cantin J, El-Baroudy S, Mejia EM, Gregorchuk B. Assessment of rapid wastewater

surveillance for determination of communicable disease spread in municipalities. *Sci. Total Environ*. 2023 Nov 25;901:166541.

16. Asadi, M., Oloye, F. F., Xie, Y., Cantin, J., Challis, J. K., McPhedran, Yusuf W, Champredon D, **Xia P**, Lange C, El-Baroudy S, Servos M, Jones P, Giesy J, Brinkmann, M. A wastewater-based risk index for SARS-CoV-2 infections among three cities on the Canadian Prairie. *Sci. Total Environ*. 2023, 876, 162800.
17. Tian M, **Xia P**, Yan L, Gou X, Yu H, Zhang X. Human functional genomics reveals toxicological mechanism underlying genotoxicants-induced inflammatory responses under low doses exposure. *Chemosphere*. 2023 Feb 1;314:137658.
18. Yan L, Gou X, **Xia P**, Gao R, Ji H, Shi W, Yu H, Zhang X. Omics-based high throughput toxicity screening and risk assessment of pollutants. *Chinese Science Bulletin.* 2022, DOI: 10.1360/TB-2022-0682.
19. Guo J, Shen Y, Zhang X, Lin D, **Xia P**, Song M, Yan L, Zhong W, Gou X, Wang C, Wei S. Effect-directed analysis based on the reduced human transcriptome (RHT) to identify organic contaminants in source and tap waters along the Yangtze River. *Environ. Sci. Technol*. 2022, May 26;56(12):7840-52. (Supplementary Cover)
20. Tian M, **Xia P**, Yan L, Gou X, Giesy J, Dai J, Yu H, Zhang X. Toxicological Mechanism of Individual Susceptibility to Formaldehyde-Induced Respiratory Effects. *Environ. Sci. Technol.* 2022, 56(10): 6511-6524. (Supplementary Cover)
21. Tian M#, **Xia P#**, Gou X, Yan L, Yu H, Zhang X. (2022) CRISPR screen identified that UGT1A9 was required for bisphenols-induced mitochondria dyshomeostasis. *Environmental Research*. 2022, 205, 112427.
22. Ha K, **Xia P**, Crump D, et al. Cytotoxic and Transcriptomic Effects in Avian Hepatocytes Exposed to a Complex Mixture from Air Samples, and Their Relation to the Organic Flame Retardant Signature. *Toxics*. 2021, 9(12): 324.
23. **Xia P**, Ying P, Fang W, Tian M, Shen Y, Ma C, Crump D, O'Brien JM, Shi W, Zhang X. (2021) Cross-Model Comparison of Transcriptomic Dose-Response of Short-Chain Chlorinated Paraffins. *Environ. Sci. Technol.* 2021, 55(12): 8149-8158.
24. Zahaby Y#, **Xia P#**, Crump D, Provencher J, Thomas P, Pauli B, Braune B, Franckowiak R, Gendron M, Savard G, Sarma S, Mallory M, and O'Brien JM. ToxChip PCR Arrays for Two Arctic-Breeding Seabirds: Applications for Regional Environmental Assessments. *Environ. Sci. Technol.* 2021, 55(11): 7521-7530.
25. Guan M, **Xia P**, Tian M, Chen D, Zhang X. Molecular fingerprints of conazoles via functional genomic profiling of Saccharomyces cerevisiae. *Toxicology in Vitro*. 2020, 104998.
26. Wang P, **Xia P**, Wang Z, Zhang X. Evidence-based assessment on environmental mixture using a concentration-dependent transcriptomics approach. *Environmental Pollution*. 2020, 114839.

27. Chen Q, Zhou C, Shi W, Wang X, **Xia P**, Song M, Liu J, Zhu H, Zhang X, Wei S, Yu H. Mechanistic in silico modeling of bisphenols to predict estrogen and glucocorticoid disrupting potentials. *Science of The Total Environment.* 2020, 138854.
28. **Xia P**, Crump D, Chiu S, Chan HM, O'Brien JM. Toxicogenomic Assessment of Complex Chemical Signatures in Double-Crested Cormorant Embryos from Variably Contaminated Great Lakes Sites. *Environ. Sci. Technol.* 2020, 54, (12), 7504-7512.
29. Fang W, Peng Y, Yan L, **Xia P**, Zhang X. A Tiered Approach for Screening and Assessment of Environmental Mixtures by Omics and In Vitro Assays. *Environ. Sci. Technol.* 2020, 54, (12), 7430-7439.
30. Wei F, Wang D, Li H, **Xia P**, Ran Y, You J. Toxicogenomics provides insights to toxicity pathways of neonicotinoids to aquatic insect, Chironomus dilutus. *Environmental Pollution*. 2020, 260, 114011.
31. Wang P, Wang Z, **Xia P**, Zhang X. Concentration-dependent transcriptome of zebrafish embryo for environmental chemical assessment. *Chemosphere*. 2020, 245, 125632.
32. Saquib Q, **Xia P**, Siddiqui MA, Zhang J, Xie Y, Faisal M, Ansari SM, Alwathnani HA, Alatar AA, Al-Khedhairy AA, Zhang X. High-throughput transcriptomics: An insight on the pathways affected in HepG2 cells exposed to nickel oxide nanoparticles. *Chemosphere*. 2020 Apr 1;244:125488.
33. **Xia P**, Zhang H, Peng Y, Shi W, Zhang X. Pathway-based assessment of single chemicals and mixtures by a high-throughput transcriptomics approach. *Environment international*. 2020 Mar 1;136:105455.
34. Xie Y, Floehr T, Zhang X, Xiao H, Yang J, **Xia P**, Burton Jr GA, Hollert H. In situ microbiota distinguished primary anthropogenic stressor in freshwater sediments. *Environmental Pollution*. 2018 Aug 1;239:189-97.
35. Zhang X, **Xia P**, Wang P, Yang J, Baird DJ. Omics Advances in Ecotoxicology. *Environ. Sci. Technol.* 2018, 52, (7), 3842-3851. (ES&T Invited Feature Article)
36. Wang P#, **Xia P#**, Yang J, Wang Z, Peng Y, Shi W, Villeneuve DL, Yu H, Zhang X. A Reduced Transcriptome Approach to Assess Environmental Toxicants Using Zebrafish Embryo Test. *Environ. Sci. Technol.* 2018, 52, (2), 821-830.
37. Wang Z, Peng Y, Wang P, **Xia P**, Zhang X. Advances of chemical testing methodologies based on zebrafish toxicogenomics. *Asian Journal of Ecotoxicology*. 2018(5):1-0.
38. Wu Y, Shi W, **Xia P**, Zhang X, Yu H. Qualitative and quantitative simulation of androgen receptor antagonists: A case study of polybrominated diphenyl ethers. *Science of The Total Environment.* 2017, 603-604, 495-501.
39. Fan M, Liu Z, Dyer S, **Xia P**, Zhang X. Environmental risk assessment of polycyclic musks HHCB and AHTN in consumer product chemicals in China. *Science of The Total Environment.* 2017, 599-600, 771-779.

40. **Xia P**, Zhang X, Zhang H, Wang P, Tian M, Yu H. Benchmarking Water Quality from Wastewater to Drinking Waters Using Reduced Transcriptome of Human Cells. *Environ. Sci. Technol.* 2017, 51, (16), 9318-9326.
41. Wang X, Zhang X, **Xia P**, Zhang J, Wang Y, Zhang R, Giesy JP, Shi W, Yu H. A high-throughput, computational system to predict if environmental contaminants can bind to human nuclear receptors. *Science of The Total Environment.* 2017, 576, 609-616.
42. Peng Y#, **Xia P#**, Zhang J, Villeneuve DL, Zhang J, Wang Z, Wei S, Yu H, Zhang X. Toxicogenomic Assessment of 6-OH-BDE47-Induced Developmental Toxicity in Chicken Embryos. *Environ. Sci. Technol.* 2016, 50, (22), 12493-12503.
43. **Xia P**, Zhang X, Xie Y, Guan M, Villeneuve DL, Yu H. Functional Toxicogenomic Assessment of Triclosan in Human HepG2 Cells Using Genome-Wide CRISPR-Cas9 Screening. *Environ. Sci. Technol.* 2016, 50, (19), 10682-10692. (Front Cover, ES&T 2016 Best Paper Award, ACS Editor' Choice)
44. Xie Y, **Xia P**, Wang H, Yu H, Giesy JP, Zhang Y, Mora MA, Zhang X. Effects of captivity and artificial breeding on microbiota in feces of the red-crowned crane (Grus japonensis). *Scientific Reports.* 2016, 6, 33350.
45. Wei F, Li J, Zhang R, **Xia P**, Peng Y, Giesy JP, Zhang X. Relative sensitivities among avian species to individual and mixtures of aryl hydrocarbon receptor-active compounds. *Environ. Toxicol. Chem.* 2016, 35, (5), 1239-46.
46. Zhang J, Zhang X, **Xia P**, Zhang R, Wu Y, Xia J, Su G, Zhang J, Giesy JP, Wang Z, Villeneuve DL, Yu H. Activation of AhR-mediated toxicity pathway by emerging pollutants polychlorinated diphenyl sulfides. *Chemosphere*. 2016, 144, 1754-62.
47. Wei F, Zhang J, **Xia P**, Zhang X, Yu H. Research progress on dioxin-like compounds and ahr-mediated adverse outcome pathway (AOP). *Asian Journal of Ecotoxicology*. 2016(1):37-51.

BOOK CHARPTERS

1. **Xia P**, Wang P, Fang W, Zhang X. Adverse Outcome Pathway Network-Based Chemical Risk Assessment Using High-Throughput Transcriptomics//Advances in Toxicology and Risk Assessment of Nanomaterials and Emerging Contaminants. Springer, Singapore, 2022: 307-324.
2. Zhang X, Wang P, **Xia P**. (2020). Dose-Dependent Transcriptomic Approach for Mechanistic Screening in Chemical Risk Assessment. In A New Paradigm for Environmental Chemistry and Toxicology (pp. 33-56). Springer, Singapore.

ISSUED PATENTS

1. Zhang X, **Xia P**. Chemical Safety Information Fast Query and Whole-Cycle Tracking Digital Label System. China (2015) Patent authorization number: cn ZL 2015 1 0220524.4.
2. Zhang X, Yang N, **Xia P**. The method and application of a primer for detecting the gene expression of enzymes related to phosphorus metabolism in houttuynia cordata. CN105219852B.
3. Zhang X, **Xia P**. Database System of Information on Environmental Safety of Chemicals. China (2015) Software registration number: cn 2015SR195803.
4. Zhang X, **Xia P**. Software of Digital Labels of Chemical Safety. China (2015) Software registration number: cn 2015SR195847.